Actuarial Decision-Support Prototype

A complete start-to-finish implementation report for the high-cost Medicare beneficiary risk

1. What Was Built

End-to-end system

2. Source Data Inventory

CMS DE-SynPUF source files are staged and standardized before aggregation to beneficiary-year gold features.

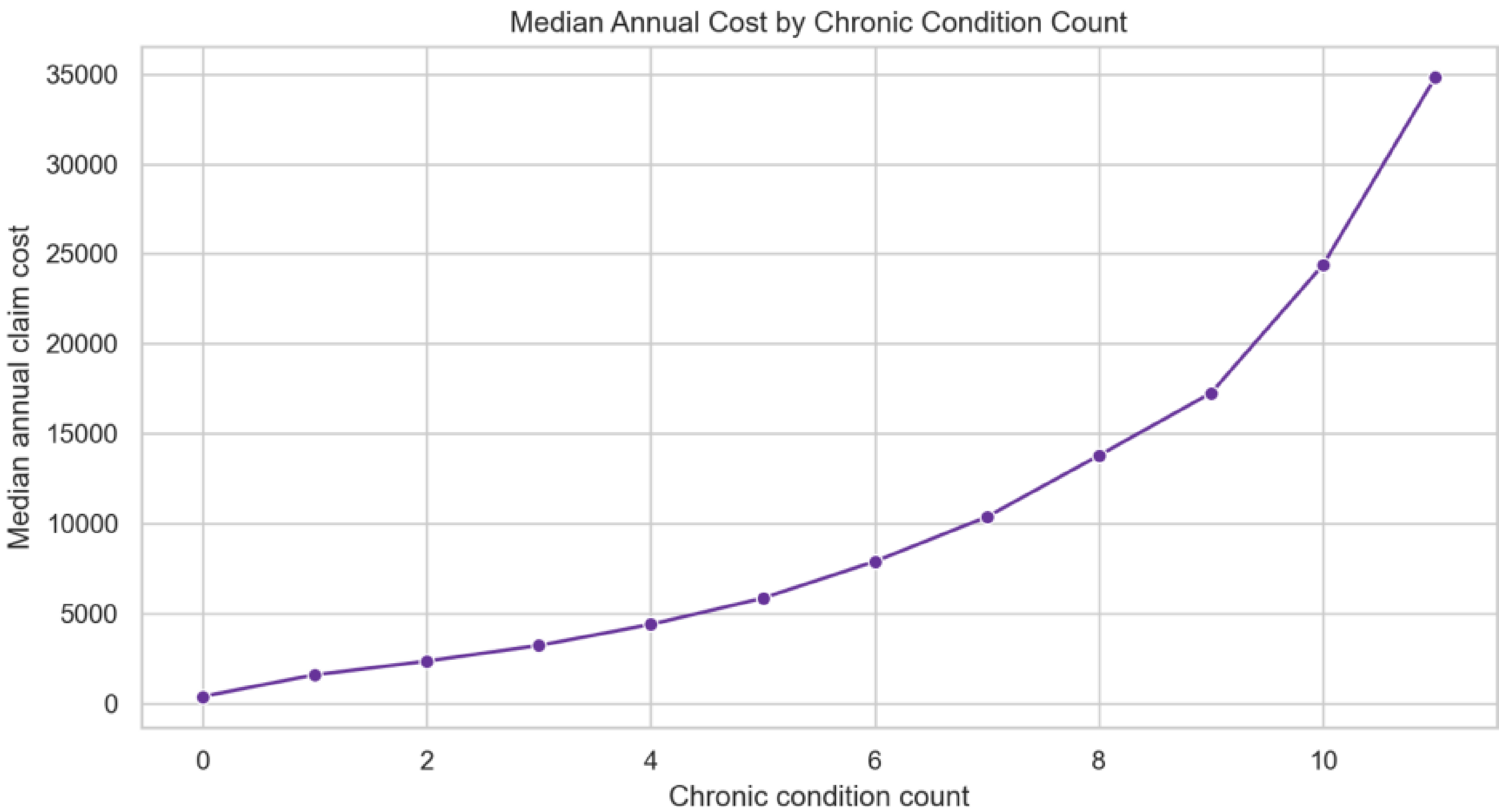
Entity	Logical source	Rows	Silver table
beneficiary_summary	beneficiary_2008	116,352	silver_beneficiaries
beneficiary_summary	beneficiary_2009	114,538	silver_beneficiaries
beneficiary_summary	beneficiary_2010	112,754	silver_beneficiaries
inpatient_claims	inpatient_2008_2010	66,773	silver_inpatient_claims
outpatient_claims	outpatient_2008_2010	790,790	silver_outpatient_claims
carrier_claims	carrier_2008_2010_a	2,370,667	silver_carrier_claims
carrier_claims	carrier_2008_2010_b	2,370,668	silver_carrier_claims
prescription_drug_events	pde_2008_2010	5,552,421	silver_pde

3. Data Pipeline and Gold Grain

Medallion architecture

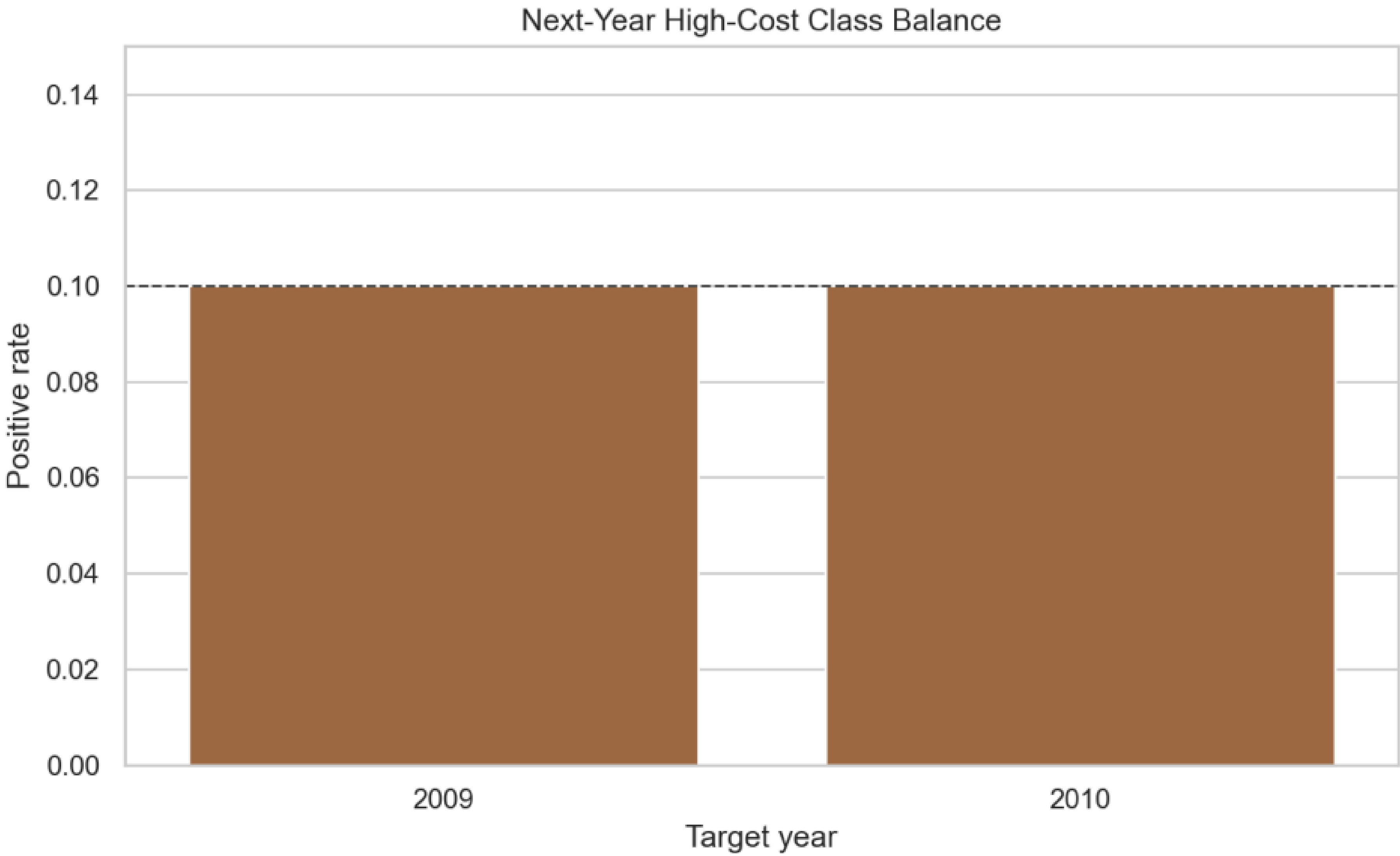
4. Chronic Burden Signal

Median annual claim cost rises with chronic-condition count, supporting chronic burden features and nonlinear terms.



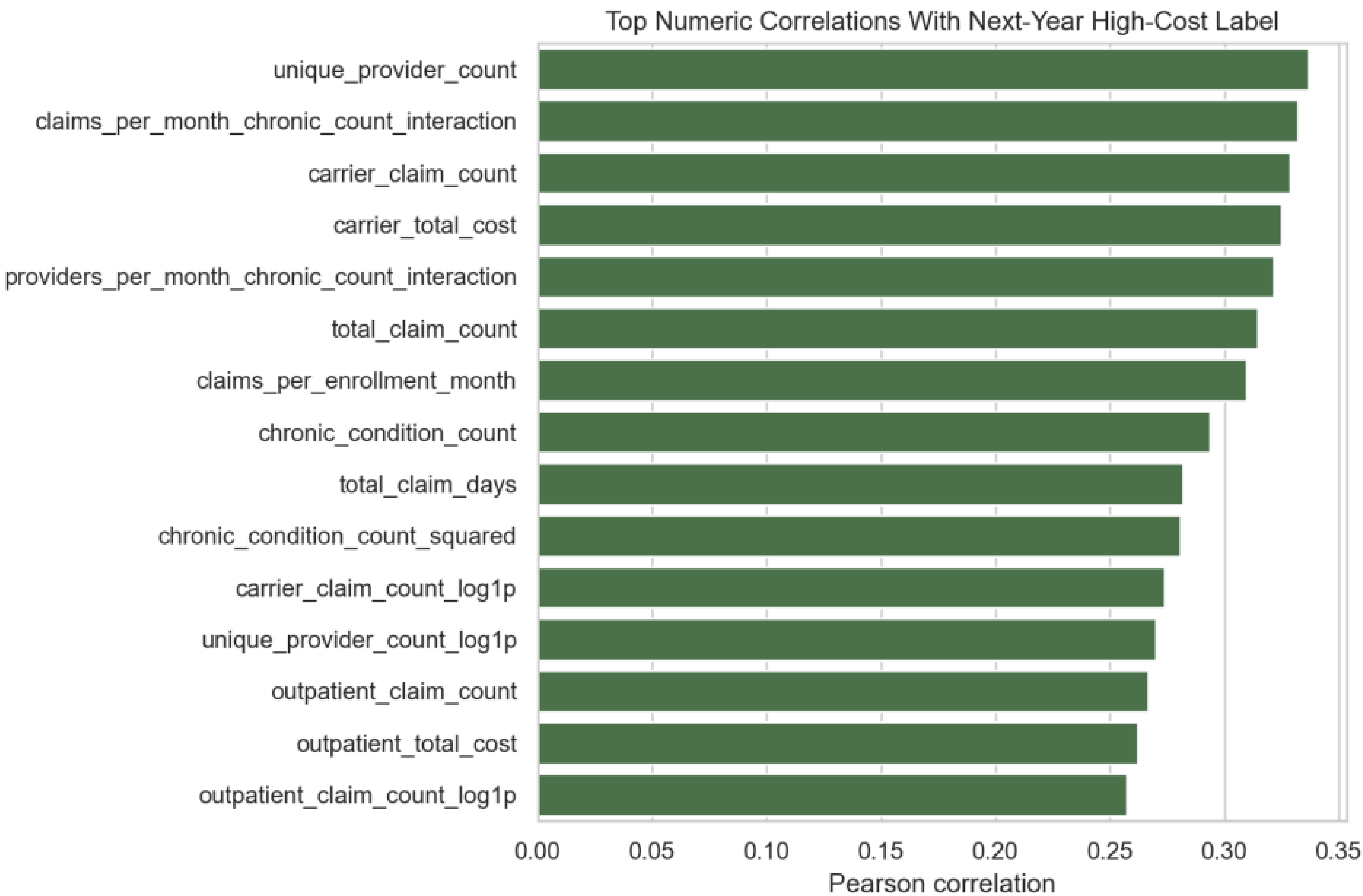
5. Target Balance by Year

The top-decile target creates an intentionally imbalanced but operationally meaningful high-cost label.



6. Feature Signal Checks

Utilization intensity, provider count, chronic burden, and cost features show the strongest prospective signal.



7. Prospective Target and Leakage Controls

Features are measured in beneficiary-year t. The high-cost label is computed from annual claim

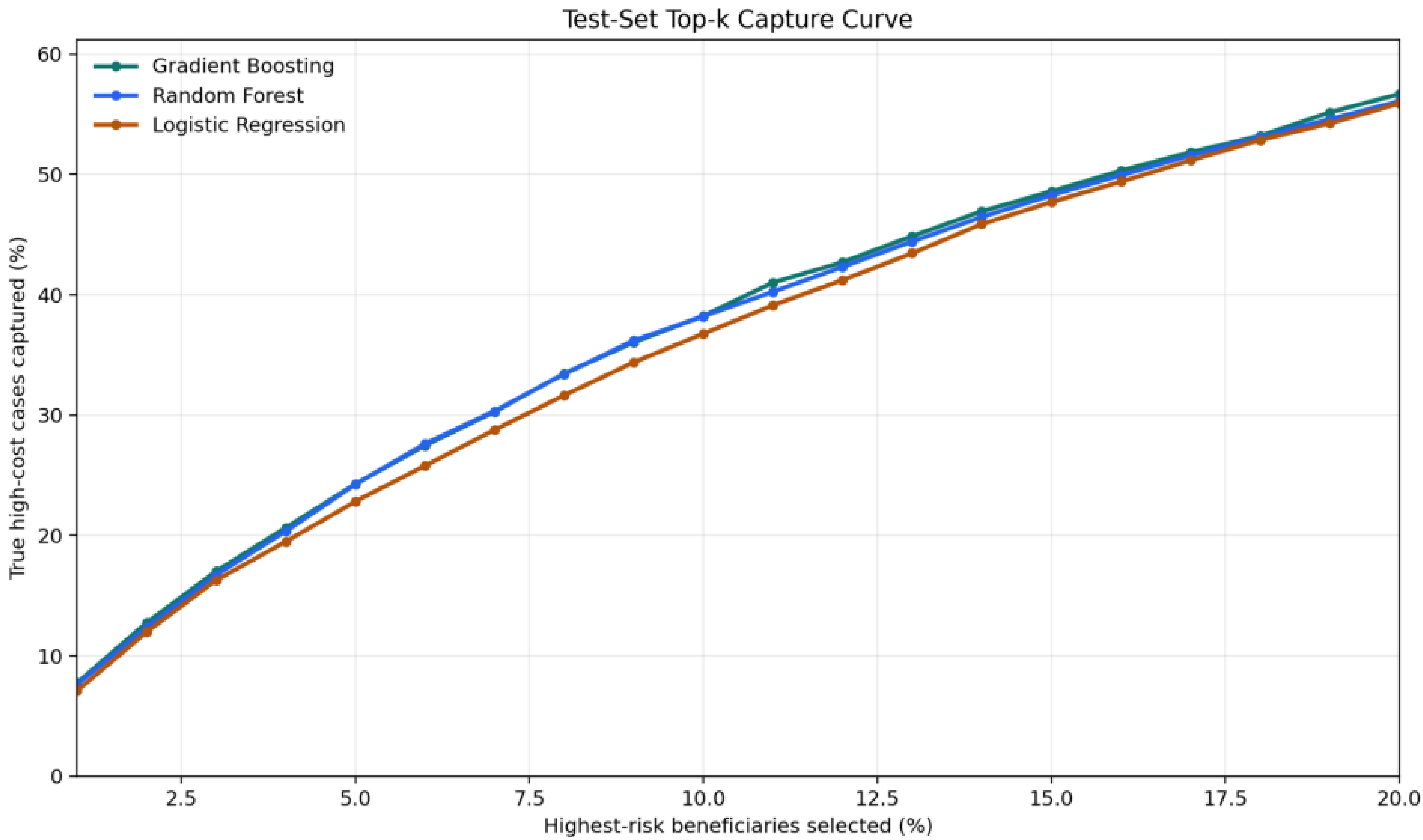
8. Final Held-Out Test Results

Gradient boosting is selected for operational ranking because it has the strongest held-out PR-AUC and top-k capture profile.

Model	Accuracy	Precision	Recall	ROC-AUC	PR-AUC	Top 5% Capture	Top 10% Capture	Top 10% Lift
Logistic Regression	0.8603	0.3432	0.4469	0.8060	0.3634	0.2386	0.3799	3.7994
Gradient Boosting	0.8533	0.3329	0.4769	0.8099	0.3879	0.2511	0.3877	3.8767
Random Forest	0.8525	0.3207	0.4747	0.8030	0.3634	0.2493	0.3819	3.8192
XGBoost	0.7131	0.2179	0.7145	0.7989	0.3617	0.2322	0.3691	3.6897

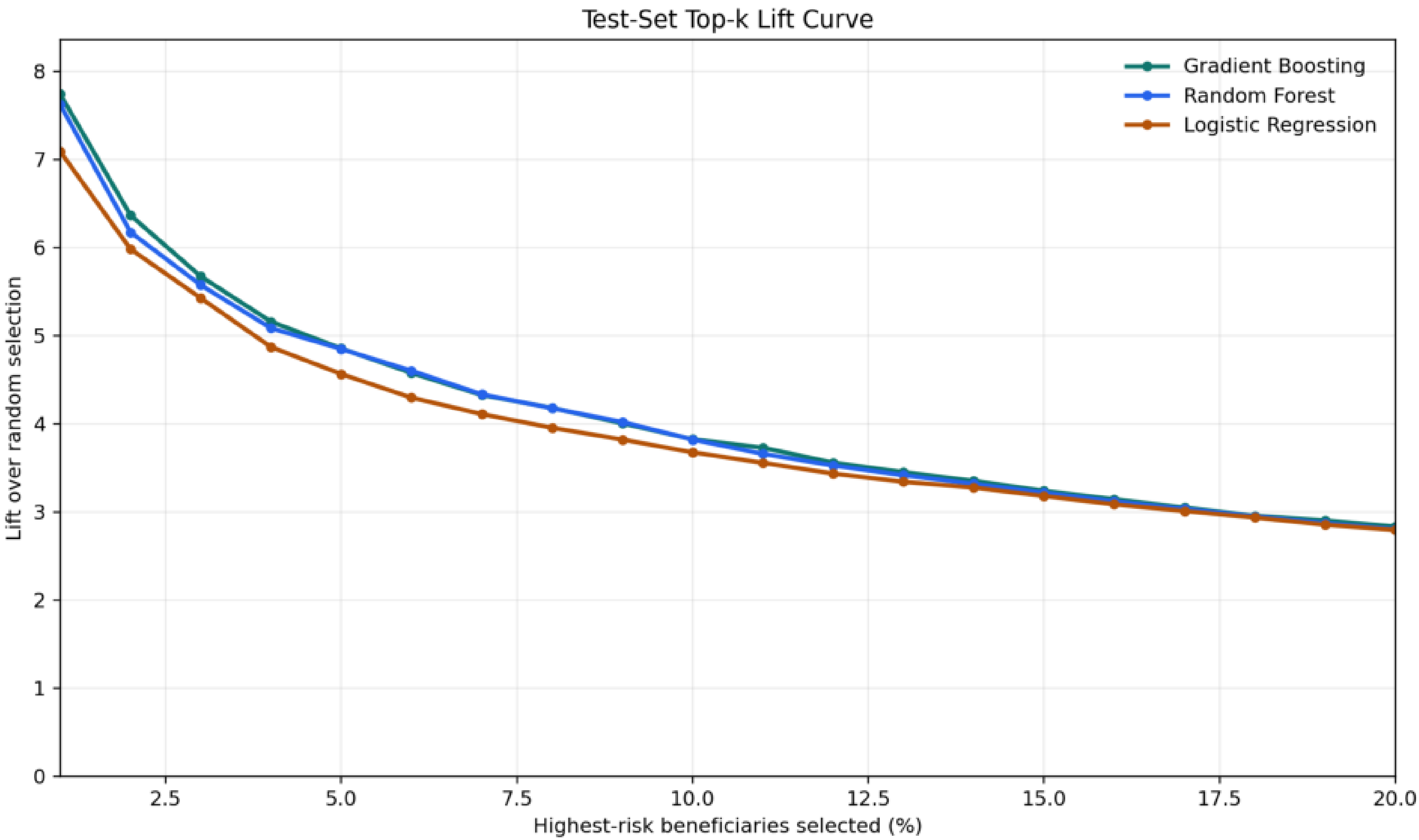
9. Top-K Capture Curve

Top-k capture shows how many true high-cost beneficiaries are captured when operations can only review the highest-risk fraction.



10. Top-K Lift Curve

Lift translates ranking quality into operational concentration versus random selection.



11. Model Selection Interpretation

- Logistic regression remains the interpretable statistical baseline.

12. API and Serving Layer

FastAPI endpoints

13. Streamlit Frontend and Demo

The frontend runs from frontend/app.py through ./run_frontend.sh and is available at

14. Simulated MDP/Q-Learning Policy Layer

The policy layer maps supervised risk into an operational MDP state.

15. Governance and Methodological Controls

- Intended use: care-management queue prioritization, actuarial risk segmentation, and

16. Verification Completed

The final local verification sequence was run after submission-control edits.

17. How to Reproduce

Local development

18. Final Takeaway

The project is a governed actuarial risk-ranking and decision-support pipeline.